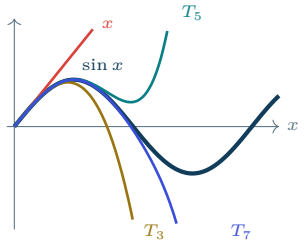
FUNCTIONS LADDER 8 OF 10

# Taylor series

A smooth function is the sum of a power series whose coefficients are its derivatives at one point.

## 1 / THE RULE AND THE PICTURE

### Matching derivatives at $0$ gives $f(x) = \sum f^{(n)}(0)\,x^n/n!$




$\sin x$ and its polynomials $x$, $T_3$, $T_5$, $T_7$

A polynomial with the same value and the first $n$ derivatives as $f$ at $0$ must have coefficients $f^{(k)}(0)/k!$:

$$f(x) = f(0) + f^{(1)}(0)x + \tfrac{f^{(2)}(0)}{2!}x^2 + \tfrac{f^{(3)}(0)}{3!}x^3 + \cdots$$

For $\sin$ the derivatives at $0$ run $0, 1, 0, -1, 0, 1, \ldots$, so $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$, and each extra term hugs the curve for longer.

## 2 / THE STANDARD SERIES

### Seven series that cover most uses

| function | series about $0$ | converges for |
|---|---|---|
| $e^x$ | $\sum_{n\ge 0} \frac{x^n}{n!}$ | all $x$ |
| $\sin x$ | $\sum_{n\ge 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ | all $x$ |
| $\cos x$ | $\sum_{n\ge 0} \frac{(-1)^n x^{2n}}{(2n)!}$ | all $x$ |
| $\frac{1}{1-x}$ | $\sum_{n\ge 0} x^n$ | $\lvert x\rvert < 1$ |
| $\ln(1+x)$ | $\sum_{n\ge 1} \frac{(-1)^{n+1} x^n}{n}$ | $-1 < x \le 1$ |
| $\arctan x$ | $\sum_{n\ge 0} \frac{(-1)^n x^{2n+1}}{2n+1}$ | $\lvert x\rvert \le 1$ |
| $(1+x)^a$ | $\sum_{n\ge 0} \binom{a}{n} x^n$ | $\lvert x\rvert < 1$ |

The binomial series generalises $(1+x)^n$ to any exponent $a$, with $\binom{a}{n} = \frac{a(a-1)\cdots(a-n+1)}{n!}$; for $a = -\frac{1}{2}$ it gives $1/\sqrt{1+x} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \cdots$.

## 3 / HOW ACCURATE

### The first omitted term gives the size of the error, for series that alternate

At $x = 1$ the sums of $\sin$ are $T_3 = 0.83333$, $T_5 = 0.84167$ and $T_7 = 0.84147$ against $\sin 1 = 0.841471$, with errors $0.0081$, $0.00020$ and $0.0000027$, each smaller than the first omitted term, $\frac{1}{120}$, $\frac{1}{5040}$ and $\frac{1}{362\,880}$.
Also $e^{0.5} = 1.64872$ against the sum of five terms $1.64844$, and $\ln 1.2 = 0.18232$ against $0.2 - 0.02 + 0.00267 - 0.0004 = 0.18227$.

## 4 / CONVERGENCE / THE RADIUS

### A series converges inside a circle in the complex plane that reaches the nearest singularity

The series for $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ fails for $\lvert x\rvert \ge 1$ although the function is smooth on the whole real line, because it has poles at $x = \pm i$, at distance $1$ from the origin, and the same series for $\arctan x$ converges only for $\lvert x\rvert \le 1$. The Leibniz series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ is the case $x = 1$ and converges slowly: ten terms give $3.0418$ and a thousand give $3.1406$.

## 5 / WHAT SERIES ARE FOR

### Limits, approximations and integrals that have no closed form

Dividing the series gives limits at once: $\frac{e^x - 1}{x} \to 1$ and $\frac{\ln(1+x)}{x} \to 1$ as $x \to 0$. Keeping two terms gives the small-angle and growth approximations, and integrating term by term handles $\int e^{-x^2}dx = x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots$, which no elementary function expresses.
Substituting $i\theta$ into the exponential series proves Euler's formula, and the ratio of successive coefficients is the first hint of the hypergeometric pattern.

### ASIDE / ANALYTIC

A function equal to its Taylor series is called analytic, and then its values anywhere are fixed by its behaviour near one point.

### LINKS TO OTHER SHEETS

The first terms are used in *Small-angle approximations*; the series for $e^x$ is in *Exponential functions and the number $e$*; the next sheet is *The gamma function and the beta function*.

**Convention.** $f^{(n)}(0)$ is the $n$th derivative at $0$; $T_n$ is the Taylor polynomial of degree $n$; $\binom{a}{n}$ is the generalised binomial coefficient.